Code No: 135CU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
NON-CONVENTIONAL POWER GENERATION
(Common to CE, ME, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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Illustrate your answers with NEAT sketches wherever necessary.

1.a) What features of solar energy make it attractive for use in irrigation water pump? With a schematic diagram, explain the working of a solar thermal water pump.
b) What is solar time, and why is it different from standard clock time of the country? Derive an expression for solar day length. [7+8]

2.a) With a schematic diagram, explain the working of a solar pond electric power plant.
b) Calculate the angle of incidence on a horizontal plane surface at Kolkata at 14.00 hrs IST on March 21st in a leap year. The north, longitude and latitude of Kolkata are $88^0\,20'$ E and $22^0\,32'$ N respectively. The standard longitude of IST is $81^0\,44'$ E. [8+7]

3.a) With a neat sketch, discuss the operating characteristics of photo voltaic module.
b) Explain the design considerations/strategies for MPPT operations. [7+8]

4.a) Explain the following systems used for converting solar radiation into electrical energy**:** (i) Central receiver system (ii) Distributed collector system.
b) Draw the equivalent circuit of a solar cell, and discuss its $V - I$ characteristics. [8+7]

5. Wind blows with a velocity of 15 m/s at 15^0C and 1 standard atm pressure. The turbine diameter is 120 m with operating speed of 40 rpm at maximum efficiency. Propeller type wind turbine is considered. Calculate the following**:** (a) total power density in the wind stream, (b) the maximum obtainable power density, (c) a reasonably obtainable power density, (d) total power, (e) torque at maximum efficiency and (f) maximum axial thrust. Assume R = 0.287 kJ/kg K and efficiency = 35 %. [15]

6.a) Explain the basic principle of wind energy conversion.
b) Discuss the merits and demerits associated with wind energy systems. [7+8]

7.a) List out various types of Geothermal resources.
b) The following data are given for a family biogas digester suitable for the output of four cows**:**
Retention time = 22 days;
Temperature = 30^0C;
Dry matter consumed = 1.8 kg/day;
Biogas yield = 0.25 m^3 per kg;
Burner efficiency = 60%;
Methane proportion = 0.82;
Heat of combustion of Methane may be assumed as 28 MJ/m^3 at STP.
Calculate the volume of the biogas digester and the power available from the digester. [6+9]

8.a) Explain the total flow concept of liquid-dominated geothermal system.
b) List the factors affecting biogas generation. Explain any four in brief. [7+8]

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